

I connected to the USASpending.gov API using a Python script. I can pull all DoD construction spending directly. This data is essentially the same as what Colby downloaded, but the API approach lets us filter to DoD before downloading, making it more efficient and replicable. I've attached a CSV showing the most relevant fields.

The data is organized around base contracts and modifications. A modification number of 0 means a single base contract with no changes. Higher numbers and letter codes indicate the modification type and sequence (e.g., B = supplemental agreement for work within scope, C = funding-only action, D = change order, K = close out). It's nice we have that level of detail.

I'm currently pulling 2020–2026, but I can expand that range. PSC codes used:

- Y – New Construction
- Z – Maintenance, Repair & Alteration
- C – Architect & Engineer (A&E) Services (not included in the CSV attached but I will add it to the python script)

Each PSC letter is followed by characters that denote facility type (e.g., Y1FC = Troop Housing). One caveat: building type classification is inconsistent — CDCs, for example, appear under several different codes.

What we can learn from the data:

1. **Spend & Cost:** Total construction and A&E spend, spend by project, spend by department, spend by facility type (via PSC mapping), spend by NAICS/PSC category, average and median award size, min/max award value
2. **Contractors & Vendors:** Top general contractors and A&E firms by total value, award count by contractor, % of total spend by top contractors, total number of contractors, average award per contractor
3. **Change Orders & Modifications:** Change order count, total cost impact of modifications, % of spend from modifications, average modification size, count and value of de-obligations, breakdown by modification/reason code
4. **Project Timeline & Execution:** DD1391 date to award time, award to construction start, construction duration, total project duration, delays and delay reasons (if supplemental data added), obligation timing by month/quarter

5. **Time-Based Trends:** Total spend by year, spend by fiscal year vs calendar year, spend by quarter and month, year-over-year change %, cumulative spend over time
6. **Geographic Distribution:** Spend by state, spend by country (CONUS vs OCONUS), spend by installation/base, % of spend by region, project count by location
7. **Project & Award Characteristics:** Total number of awards, total number of projects, spend per project, number of modifications per award, distribution of award sizes (e.g., <\$1M, \$1–10M, \$10M+)
8. **Department & Agency Breakdown:** Spend by service branch (Army, Navy, Air Force, etc.), spend by awarding agency, spend by funding agency, % of total spend by department
9. **Facility & Construction Types:** Spend by facility type (barracks, CDCs, hospitals, etc.), new construction vs renovation, project count by facility type (requires mapping table)

If we add the MILCON status reports + MILCON schedule delay data + Suruchi's key dates:

1. **Schedule Performance & Delays:** Planned vs actual completion date, total delay duration (days/months), % of projects delayed (>1 year), delay duration by project, delay duration by department, delay duration by facility type, schedule variance (planned vs actual), construction percent complete vs time elapsed
2. **Delay Drivers & Root Causes:** Delay count by reason (unforeseen conditions, contractor performance, planning issues, supply chain, approvals, design errors), % of delays by category, average delay duration by reason, projects with multiple delay causes
3. **Execution & Productivity Metrics:** Planned vs actual construction pace (% complete per month), schedule slippage over time, projects ahead/on-time/behind schedule, contractor-specific schedule performance, rework-driven delays, modification-driven time extensions
4. **Pre-Construction & Planning Metrics:** Time from DD1391 to award, time from award to construction start, permitting/approval duration, design duration, % of delays attributed to planning/design issues

Here's what I propose we do:

Step 1 – Build the Crosswalk Table (requires some manual work)

I'll develop a crosswalk of historical project IDs, GC contract awards, and A&E awards so we can link everything to a single project. The GC and A&E are under separate award numbers, so this mapping step is essential. I'll run it through AI first but expect some manual cleanup to confirm matches.

Step 2 – Enrich with Schedule & Delay Data (future phase)

Attachments

- MILCON Status Report (December 2025) <https://www.acq.osd.mil/eie/imr/mc/library.html>
- CSV field format example (2020–2026 USASpending pull, this is the python output)
- MILCON Schedule Delays PDF <https://www.acq.osd.mil/eie/imr/mc/library.html>